HARD · ADVANCED MATH

Nonlinear functions

08

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years, t, it will take for the population of the town to reach 60,000 ?

A. $50{,}000 = 60{,}000(0.03)^t$

B. $50{,}000 = 60{,}000(3)^t$

C. $60{,}000 = 50{,}000(0.03)^t$

D. $60{,}000 = 50{,}000(1.03)^t$

Q2 ADVANCED MATH

$$fx = 5{,}4700.64^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p?

A. 4

B. 5

C. 36

D. 64

Q3 GRID-IN ADVANCED MATH

Function f is defined by $f x = -a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = f x - 12$ has a y-intercept at $0, -\frac{75}{7}$. The product of a and b is $\frac{320}{7}$. What is the value of a?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ADVANCED MATH

When the quadratic function f is graphed in the xy-plane, where $y = fx$, its vertex is $-3, 6$. One of the x-intercepts of this graph is $-\frac{17}{4}, 0$. What is the other x-intercept of the graph?

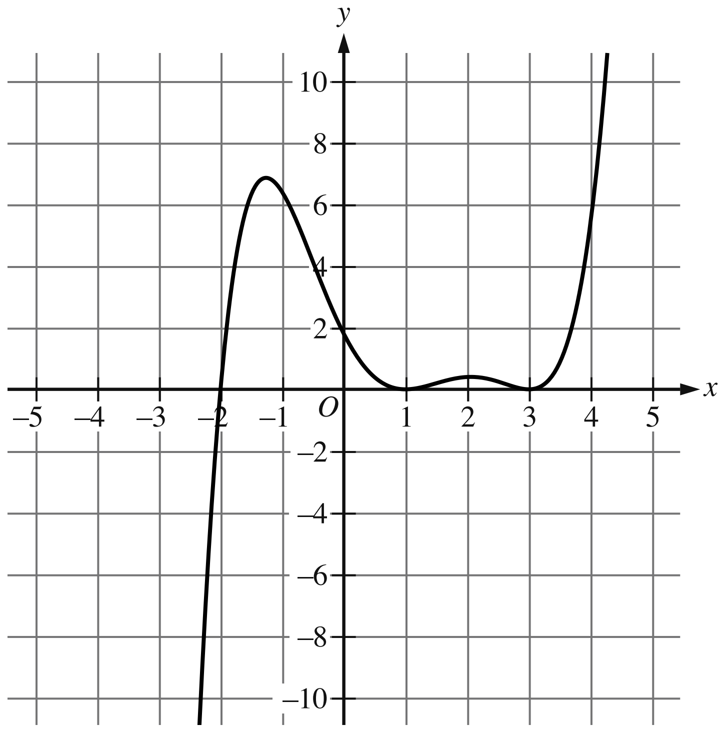
A. $-\frac{29}{4}, 0$

B. $-\frac{7}{4}, 0$

C. $\frac{5}{4}, 0$

D. $\frac{17}{4}, 0$

Q5 ADVANCED MATH



- Moving from left to right, the curve passes from quadrant 3 to quadrant 2 to quadrant 1.
- The curve has 2 relative maximums and 2 relative minimums.
- The curve crosses the x axis at the point (negative 2 comma 0).
- The curve touches the x axis at the following points:
  - (1 comma 0)
  - (3 comma 0)

The graph of $y = gx$ is shown. If $gx + 28 = \frac{1}{10}x - a^2x - b^2x - c$, where a, b, and c are constants, which of the following could be the value of b?

A. 31

B. 3

C. -2

D. -25

Q6

ADVANCED MATH

Function f is defined by $fx = x + 6x + 5x + 1$. Function g is defined by $gx = fx - 1$. The graph of $y = gx$ in the xy-plane has x-intercepts at $a, 0$, $b, 0$, and $c, 0$, where a, b, and c are distinct constants. What is the value of $a + b + c$?

A. -15

B. -9

C. 11

D. 15

Q7

GRID-IN ADVANCED MATH

The functions f and g are defined by the given equations.

$$fx = 3 + | -2x - x^2|$$

$$gw = | \frac{-w}{w-1} | - w + 5$$

If $f(-4) = c$, where c is a constant, what is the value of gc?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ADVANCED MATH

A model estimates that at the end of each year from 2015 to 2020, the number of squirrels in a population was 150% more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of 2016, there were 180 squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of 2015 and $t \leq 5$?

A. $n = 721.5^t$

B. $n = 722.5^t$

C. $n = 1801.5^t$

D. $n = 1802.5^t$

Q9 GRID-IN ADVANCED MATH

The function f is defined by $f x = x - 6x - 2x + 6$. In the xy-plane, the graph of $y = g x$ is the result of translating the graph of $y = f x$ up 4 units. What is the value of $g 0$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

$$ft = 55t - 2t^2$$

The function f is defined by the given equation. The function g is defined by $gt = ft + 3$. Which expression represents the maximum value of gt?

A. $3 + \frac{55}{2}^2$

B. $3 + 2\frac{55}{4}^2$

C. $3 - 2\frac{55}{4}^2$

D. $3 - \frac{55}{2}^2$